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1. INTRODUCTION

I made four applications for this assignment using **python and C#**. I developed a file extension information system and a contact book in C#. Then, I developed a python client-server program to monitor student admission and a web scraping program for hotels that outputs data to a CSV file. I was able to gain experience with data manipulation, programming and problem solving through these assignments.

2. C# PROGRAMMING

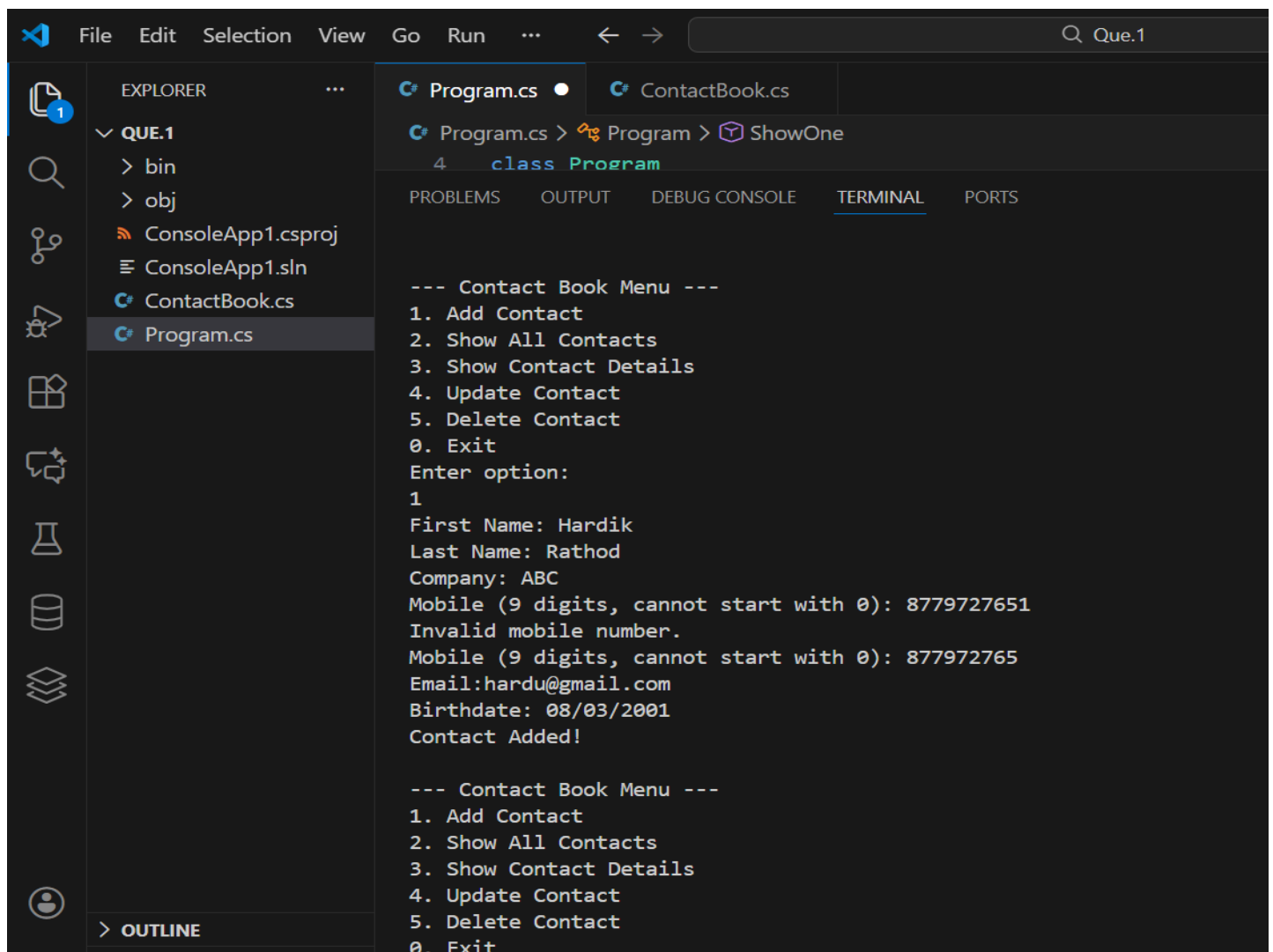
Que.1 Contact Book

For this program, I developed a C# contact book program.

This project contains two files:

- i. **Program.cs**
- ii. **ContactBook.cs**

I made a **ContactBook** class with the name, company, phone number, email, address and birthdate of every contact. At the start of the program, the array has **20 contacts**. The menu lets the user add, view, update and remove contacts while confirming that the phone number is nine digits long. I used basic OOP concepts to test the program using dotnet run, which runs continuously until the user exits.



```
File Edit Selection View Go Run ... < > Que.1

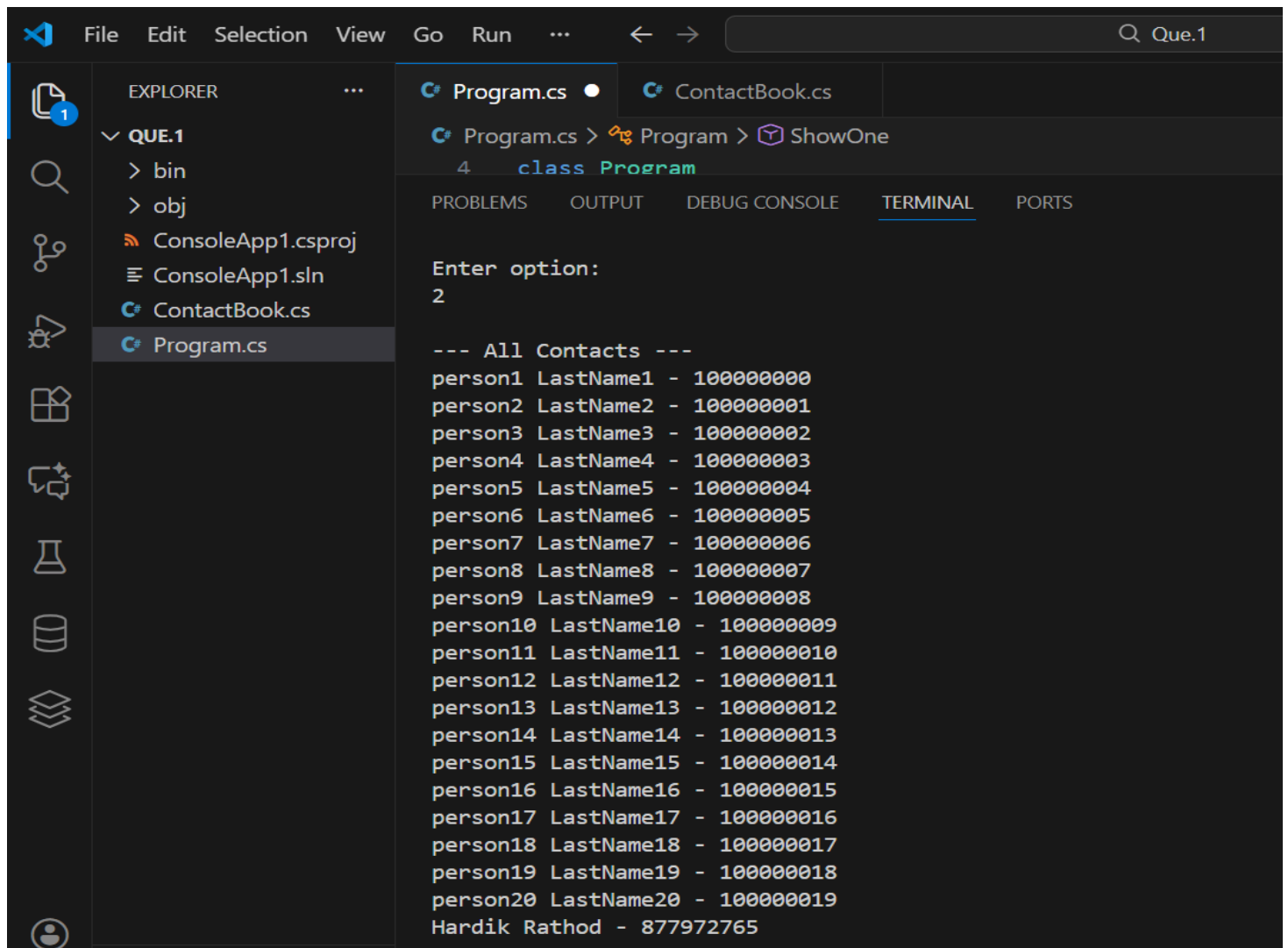
EXPLORER
  QUE.1
    > bin
    > obj
    ConsoleApp1.csproj
    ConsoleApp1.sln
    ContactBook.cs
    Program.cs

Program.cs
  Program.cs > Program > ShowOne
  4 class Program

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

--- Contact Book Menu ---
1. Add Contact
2. Show All Contacts
3. Show Contact Details
4. Update Contact
5. Delete Contact
0. Exit
Enter option:
1
First Name: Hardik
Last Name: Rathod
Company: ABC
Mobile (9 digits, cannot start with 0): 8779727651
Invalid mobile number.
Mobile (9 digits, cannot start with 0): 877972765
Email:hardu@gmail.com
Birthdate: 08/03/2001
Contact Added!

--- Contact Book Menu ---
1. Add Contact
2. Show All Contacts
3. Show Contact Details
4. Update Contact
5. Delete Contact
0. Exit
```



```
File Edit Selection View Go Run ... < > Que.1
```

EXPLORER

- QUE.1
 - > bin
 - > obj
 - ConsoleApp1.csproj
 - ConsoleApp1.sln
 - ContactBook.cs
 - Program.cs

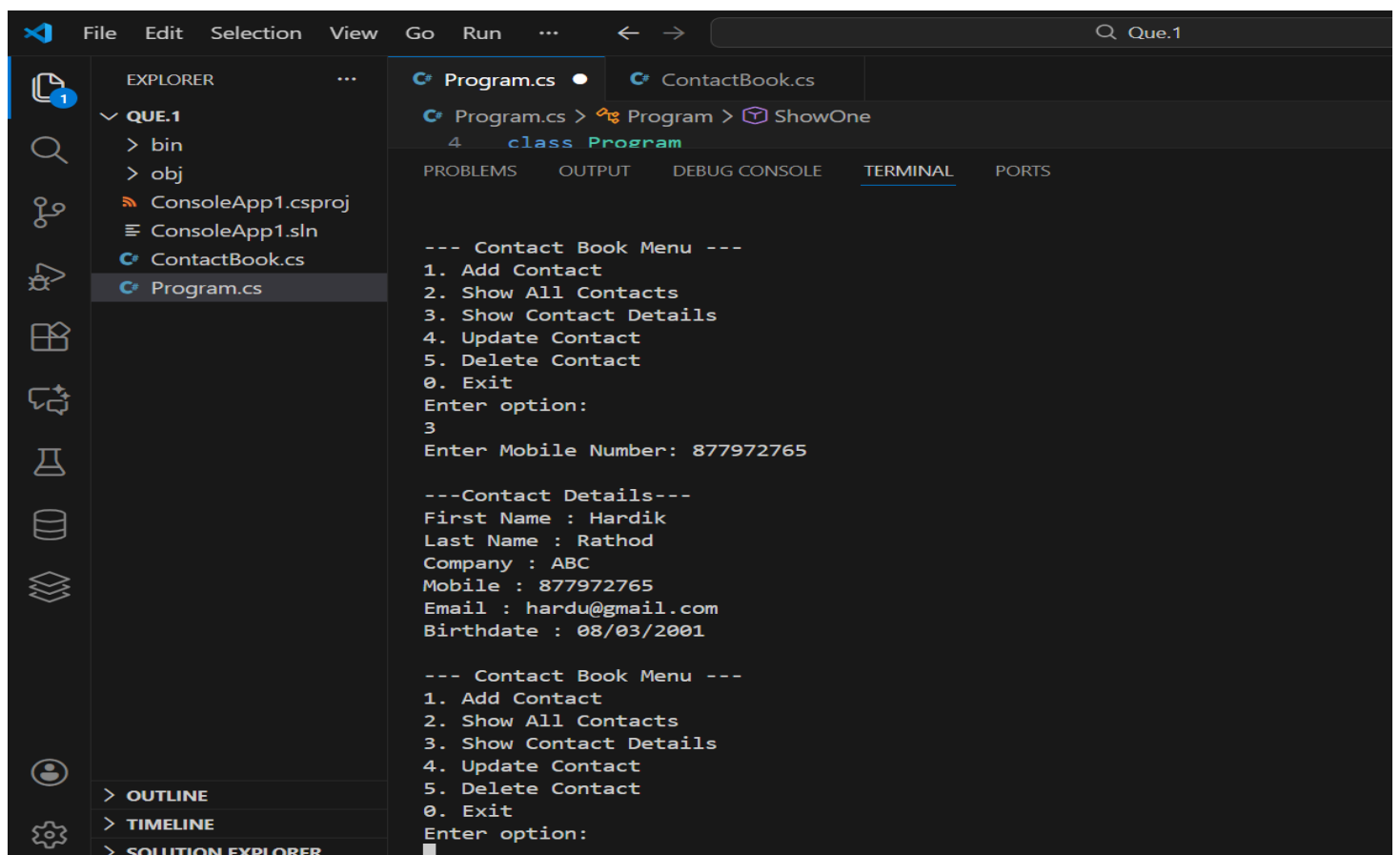
Program.cs

```
class Program
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Enter option:
2

--- All Contacts ---
person1 LastName1 - 100000000
person2 LastName2 - 100000001
person3 LastName3 - 100000002
person4 LastName4 - 100000003
person5 LastName5 - 100000004
person6 LastName6 - 100000005
person7 LastName7 - 100000006
person8 LastName8 - 100000007
person9 LastName9 - 100000008
person10 LastName10 - 100000009
person11 LastName11 - 100000010
person12 LastName12 - 100000011
person13 LastName13 - 100000012
person14 LastName14 - 100000013
person15 LastName15 - 100000014
person16 LastName16 - 100000015
person17 LastName17 - 100000016
person18 LastName18 - 100000017
person19 LastName19 - 100000018
person20 LastName20 - 100000019
Hardik Rathod - 877972765
```



```
File Edit Selection View Go Run ... < > Que.1
```

EXPLORER

- QUE.1
 - > bin
 - > obj
 - ConsoleApp1.csproj
 - ConsoleApp1.sln
 - ContactBook.cs
 - Program.cs

Program.cs

```
class Program
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
--- Contact Book Menu ---
1. Add Contact
2. Show All Contacts
3. Show Contact Details
4. Update Contact
5. Delete Contact
0. Exit
Enter option:
3
Enter Mobile Number: 877972765

---Contact Details---
First Name : Hardik
Last Name : Rathod
Company : ABC
Mobile : 877972765
Email : hardu@gmail.com
Birthdate : 08/03/2001

--- Contact Book Menu ---
1. Add Contact
2. Show All Contacts
3. Show Contact Details
4. Update Contact
5. Delete Contact
0. Exit
Enter option:
```

> OUTLINE
> TIMELINE
> SOLUTION EXPLORER

File Edit Selection View Go Run ...

EXPLORER

- QUE.1
 - > bin
 - > obj
 - ConsoleApp1.csproj
 - ConsoleApp1.sln
 - ContactBook.cs
 - Program.cs
- OUTLINE
- TIMELINE
- SOLUTION EXPLORER

Program.cs

Program.cs > Program > ShowOne

```
4 class Program
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Enter Mobile Number to Update: 932733778
Contact not found.

--- Contact Book Menu ---

1. Add Contact
2. Show All Contacts
3. Show Contact Details
4. Update Contact
5. Delete Contact
0. Exit

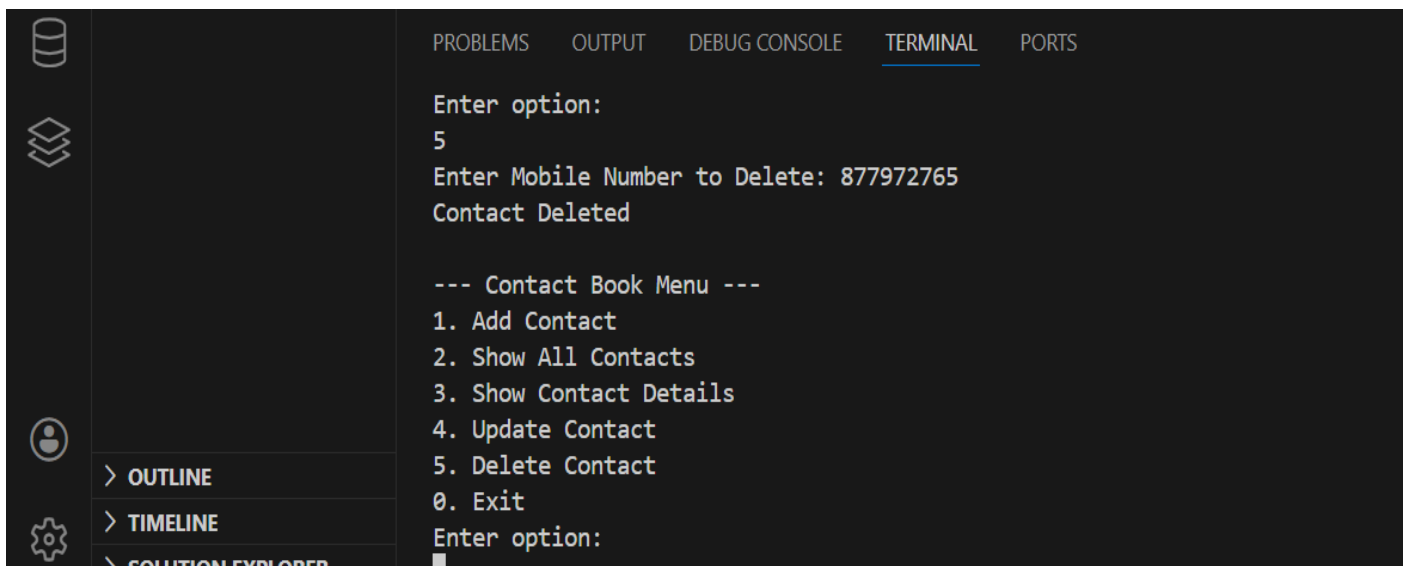
Enter option:
4

Enter Mobile Number to Update: 877972765
New First Name (Enter to skip): Rathod
New Last Name (Enter to skip): Hardik
New Company (Enter to skip): CBA
New Email (Enter to skip): rathodhardu@gmail.com
New Birthdate (Enter to skip): 08/03/2001
Contact Updated!

--- Contact Book Menu ---

1. Add Contact
2. Show All Contacts
3. Show Contact Details
4. Update Contact
5. Delete Contact
0. Exit

Enter option:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Enter option:
5
Enter Mobile Number to Delete: 877972765
Contact Deleted

--- Contact Book Menu ---
1. Add Contact
2. Show All Contacts
3. Show Contact Details
4. Update Contact
5. Delete Contact
0. Exit
Enter option:
```

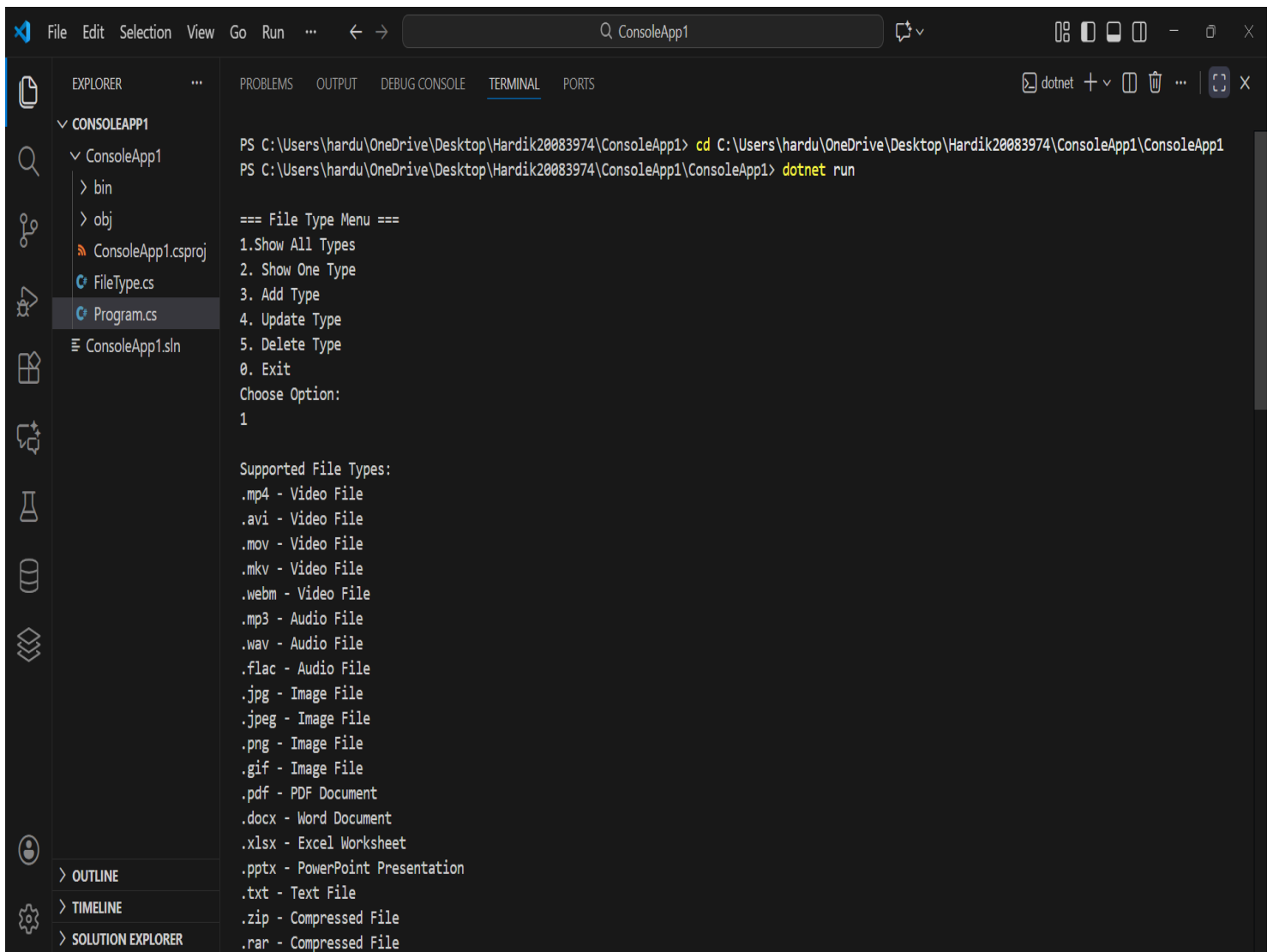
Que.2 File Extension System

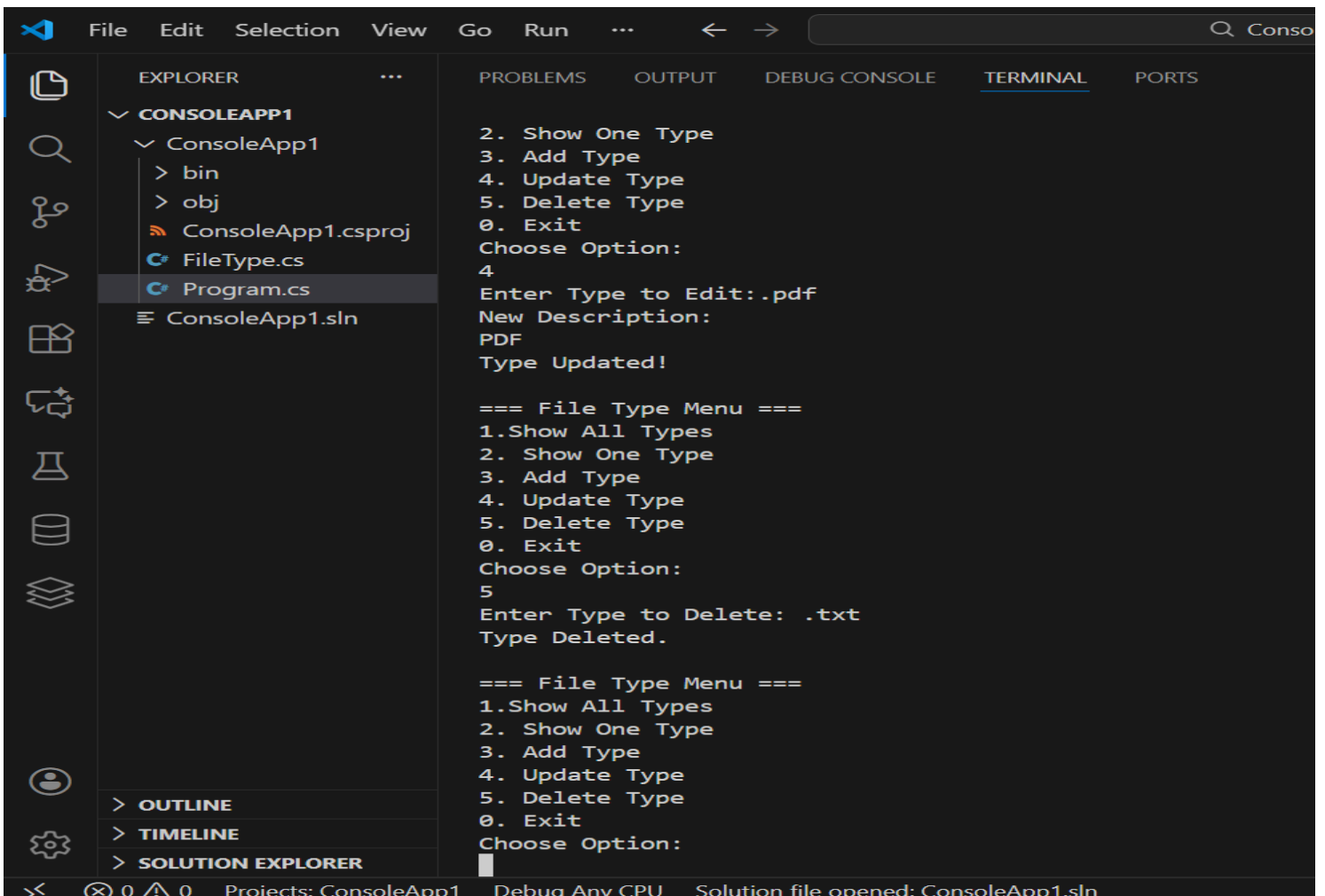
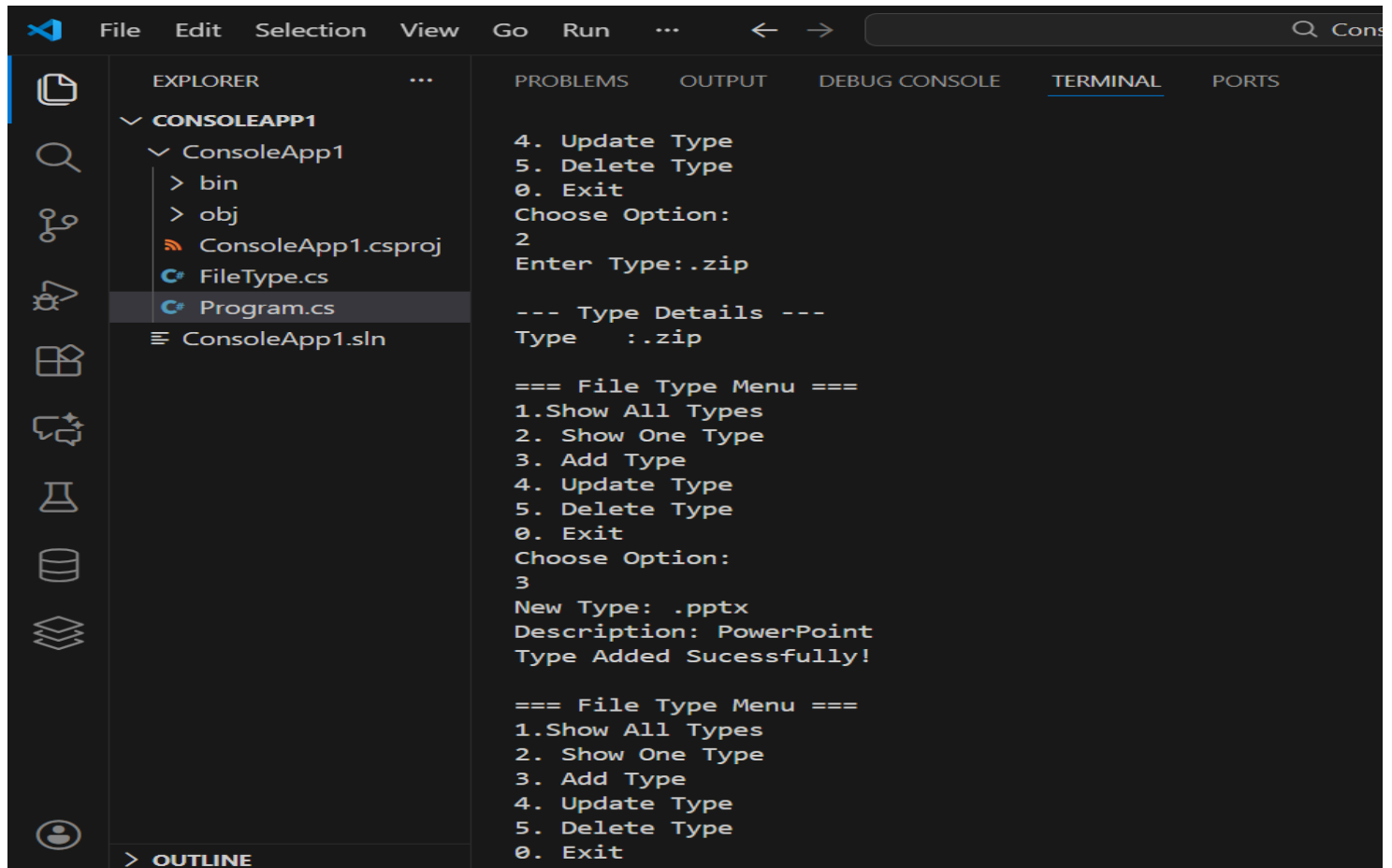
To answer this question, I created a C# console application that displays information about different file extensions

I made 2 files in this program:

- i. **Program.cs**
- ii. **FileType.cs**

I created a program that generated 20 default file extensions using the **DefaultExtention() method**. The menu allows the user to view, add, remove, update or search an extension. If the user enters an extension but forgets to do so, the program will automatically add a dot. An error message shows up if there is not an extension. I was able to use dotnet run to debug each of the fundamental OOP concepts, such as classes, objects and methods.





3. PYTHON PROGRAMMING

Que.3 Client-Server Application

Using python sockets, I created a server program that waits for a client. I created a client program that sends information about students admissions to the server. I used SQLite to create a database file that contains all of the student information. After receiving the data, the server generates and saves a unique application number. The client receives the application number from the server. The application number is sent back to the client by the server. The application number is given to the user by the client. Databases, TCP communication and basic string manipulation are used in this work.

```
client.py
1 import socket

server.py

Terminal 1:
PS C:\Users\hardu\OneDrive\Desktop\Hardik20083974\Que.3> python server.py
Server Is Running... Waiting For Incoming Connection...
Client Connected: ('127.0.0.1', 52150)
Client Connected: ('127.0.0.1', 61899)
Client Connected: ('127.0.0.1', 61906)

Terminal 2:
Start Month: January
YOUR APPLICATION ID IS: ADM5882
PS C:\Users\hardu\OneDrive\Desktop\Hardik20083974\Que.3> python client.py
Please Enter Your Details Below:
Full Name: Sakshi Teli
Address: Pune
Qualification: BCA
Course: Cyber Security
Start Year: 2025
Start Month: September

YOUR APPLICATION ID IS: ADM5833
PS C:\Users\hardu\OneDrive\Desktop\Hardik20083974\Que.3> python client.py
Please Enter Your Details Below:
Full Name: Mayuri Mistry
Address: Mumbai
Qualification: Teacher
Course: Advance Programming
Start Year: 2010
Start Month: March

YOUR APPLICATION ID IS: ADM6231
PS C:\Users\hardu\OneDrive\Desktop\Hardik20083974\Que.3>
```

The screenshot shows a VS Code editor with a SQLite database viewer and a terminal. The database viewer displays the 'STUDENTS' table with 6 rows. The terminal shows the execution of a Python server and client, with the client providing details for a student named Mayuri Mistry.

AppID	Name	Address	Qualificat...	Course	Year	Month
1 ADM3399	Hardik Rathod	Dublin	Qualification:	Course:	Start Year:	Start Month:
2 ADM2622	Hardik Rathod	Mumbai	BCA	Cyber Security	2025	September
3 ADM3924	Hardik Rathod	Dublin	BCA	Cyber Security	2025	September
4 ADM5882	Reeta Rathod	Mumbai	12th	Commerce	1980	January
5 ADM5833	Sakshi Teli	Pune	BCA	Cyber Security	2025	September
6 ADM6231	Mayuri Mistry	Mumbai	Teacher	Advance Programming	2010	March
7	Mayuri Mistry					

```

PS C:\Users\hardu\OneDrive\Desktop\Hardik20083974\Que.3> python server.py
Server Is Running... Waiting For Incoming Connection...
Client Connected: ('127.0.0.1', 52150)
Client Connected: ('127.0.0.1', 61899)
Client Connected: ('127.0.0.1', 61906)
[]

YOUR APPLICATION ID IS: ADM5833
PS C:\Users\hardu\OneDrive\Desktop\Hardik20083974\Que.3> python client.py
Please Enter Your Details Below:
Full Name: Mayuri Mistry
Address: Mumbai
Qualification: Teacher
Course: Advance Programming
Start Year: 2010
Start Month: March

YOUR APPLICATION ID IS: ADM6231
PS C:\Users\hardu\OneDrive\Desktop\Hardik20083974\Que.3>

```

Que.4 Web Scraping and CSV Handling

Since actual hotel websites do not permit web scraping for this task, I made two demo hotel webpages to practice data extraction. Since every demo website was constructed differently, I gathered the room details using two different approaches. Since JavaScript was used to generate the rooms on the first website (Hotel A), the room details are not directly displayed in the HTML.

I opened the HTML page and saved it as plain text and used the simplest form of splitting string to find the room list inside the JavaScript code since BeautifulSoup has no idea that something created by JavaScript exists. I then extracted the names and prices of the rooms. I applied BeautifulSoup to spend a few seconds in searching all the room cards in the second site (Hotel B) and extract the name and the price since the information about the rooms was already included in the HTML. I copied all the data on the two demo websites in a CSV file and printed the results to ensure that the scraping worked.

```
1 from bs4 import BeautifulSoup
2 import csv
3
4 data = []
5
6
7 with open("hotelA.html", "r", encoding="utf-8") as f:
```

```
PS C:\Users\hardu\OneDrive\Desktop\Hardik20083974\Que.4> python Que4.py
{'Hotel': 'Hotel A', 'Room': 'Deluxe Room', 'Price': '120'}
{'Hotel': 'Hotel A', 'Room': 'Standard Room', 'Price': '80'}
{'Hotel': 'Hotel A', 'Room': 'Suite Room', 'Price': '200'}
{'Hotel': 'Hotel A', 'Room': 'Single Room', 'Price': '60'}
{'Hotel': 'Hotel A', 'Room': 'Double Room', 'Price': '100'}
{'Hotel': 'Hotel A', 'Room': 'Family Room', 'Price': '180'}
{'Hotel': 'Hotel A', 'Room': 'Presidential Suite', 'Price': '350'}
{'Hotel': 'Hotel A', 'Room': 'Economy Room', 'Price': '50'}
{'Hotel': 'Hotel B', 'Room': 'Presidential Room', 'Price': '$400 /night'}
{'Hotel': 'Hotel B', 'Room': 'King Suite', 'Price': '$320 /night'}
{'Hotel': 'Hotel B', 'Room': 'Luxury Room', 'Price': '$300 /night'}
{'Hotel': 'Hotel B', 'Room': 'Standard Room', 'Price': '$90 /night'}
{'Hotel': 'Hotel B', 'Room': 'Family Suite', 'Price': '$250 /night'}
{'Hotel': 'Hotel B', 'Room': 'Single Room', 'Price': '$65 /night'}
{'Hotel': 'Hotel B', 'Room': 'Double Room', 'Price': '$110 /night'}
{'Hotel': 'Hotel B', 'Room': 'Queen Suite', 'Price': '$180 /night'}
{'Hotel': 'Hotel B', 'Room': 'Suite Room', 'Price': '$210 /night'}
{'Hotel': 'Hotel B', 'Room': 'Economy Room', 'Price': '$70 /night'}
PS C:\Users\hardu\OneDrive\Desktop\Hardik20083974\Que.4>
```

```
1 Hotel,Room,Price
2 Hotel A,Deluxe Room,120
3 Hotel A,Standard Room,80
4 Hotel A,Suite Room,200
5 Hotel A,Single Room,60
6 Hotel A,Double Room,100
7 Hotel A,Family Room,180
8 Hotel A,Presidential Suite,350
9 Hotel A,Economy Room,50
10 Hotel B,Presidential Room,$400 /night
11 Hotel B,King Suite,$320 /night
12 Hotel B,Luxury Room,$300 /night
13 Hotel B,Standard Room,$90 /night
14 Hotel B,Family Suite,$250 /night
15 Hotel B,Single Room,$65 /night
16 Hotel B,Double Room,$110 /night
17 Hotel B,Queen Suite,$180 /night
18 Hotel B,Suite Room,$210 /night
19 Hotel B,Economy Room,$70 /night
20
```

4. CONCLUSION

I developed four Python and C Sharp applications, the development of which enhanced my knowledge of OOP, data processing, databases, networking, and CSV, and raised my confidence in my programming.

5. REFERENCES

Mitchell, R. (2018). *Web Scraping with Python: Collecting data from the modern web* (2nd ed.). O'Reilly Media. <https://www.oreilly.com/library/view/web-scraping-with/9781491985564/>

Richardson, L. (2022). *Beautiful Soup documentation*.
<https://www.crummy.com/software/BeautifulSoup/>

Python Software Foundation. (2024). *Python 3 documentation*.
<https://docs.python.org/3/>

Microsoft. (2024). *Classes and objects – C# documentation*.
<https://learn.microsoft.com/en-us/dotnet/csharp/fundamentals/object-oriented/classes>

6. GITHUB REPOSITORY LINK

<https://github.com/hardurathod/HardikRathod-20083974---AdvProgramming-CA1-.git>